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To cite this article: Claudia Ventura, Andrew J. Guerin, Osama El-Zubir, Antonio J. Ruiz-Sanchez, Luke I. Dixon, Kevin J. Reynolds, Marie L. Dale, James Ferguson, Andrew Houlton, Benjamin R. Horrocks, Anthony S. Clare & David A. Fulton (2017) Marine antifouling performance of polymer coatings incorporating zwitterions, *Biofouling*, 33:10, 892-903, DOI: [10.1080/08927014.2017.1383983](https://doi.org/10.1080/08927014.2017.1383983)

To link to this article: <https://doi.org/10.1080/08927014.2017.1383983>



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Marine antifouling performance of polymer coatings incorporating zwitterions

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ABSTRACT

Zwitterionic materials display antifouling promise, but their potential in marine anti-biofouling is still largely unexplored. This study evaluates the effectiveness of incorporating small quantities (0–20% on a molar basis) of zwitterions as sulfobetaine methacrylate (SBMA) or carboxybetaine methacrylate (CBMA) into lauryl methacrylate-based coatings whose relatively hydrophobic nature encourages adhesion of the diatom *Navicula incerta*, a common microfouling organism responsible for the formation of 'slime'. This approach allows potential enhancements in antifouling afforded by zwitterion incorporation to be easily quantified. The results suggest that the incorporation of CBMA does provide a relatively minor enhancement in fouling-release performance, in contrast to SBMA which does not display any enhancement. Studies with coatings incorporating mixtures of varying ratios of the cationic monomer [2-(methacryloyloxy)ethyl]trimethylammonium chloride and the anionic monomer (3-sulfopropyl)methacrylate, which offer a potentially lower cost approach to the incorporation of anionic and cationic charge, suggest these monomers impart little significant effect on biofouling.

ARTICLE HISTORY

Received 16 February 2017
Accepted 18 September 2017

KEYWORDS

Zwitterions; polymer coatings; foul-release; *Navicula incerta*

Introduction

Marine biofouling, which can be defined as the accumulation of microorganisms, plants and animals on surfaces immersed in seawater, is an extensive problem for maritime industries. For example, increased hull roughness resulting from biofouling can increase fuel consumption of marine vessels by 40% at cruising speed, and overall voyage costs by 77% (Schultz 2007; Schultz et al. 2011, 2015). Biofouling is also a potential problem for aquaculture (Braithwaite and McEvoy 2004), offshore oil and gas production (Edyvean 1987), underwater sensors (Delauney et al. 2010) and other artificial structures in the marine environment.

Antifouling (AF) coatings are essential for the prevention of biofouling on immersed mobile and stationary structures. The most commonly used AF technologies at present are 'biocidal' coatings. These coatings rely on the controlled release of chemically active compounds, typically organometals or cuprous oxides, embedded within a polymer matrix binder. The most advanced biocidal AF coatings are based upon a self-polishing strategy, consisting of copolymers that are easily hydrolysable in seawater,

resulting in the continuous release of both biocide and binder from the surface of the coating (Chambers et al. 2006). The performance of such coatings is offset by the environmental impact of the continuous release of toxic compounds, leading to the use of tetra-butyl-tin based coatings being banned (Terlizzi et al. 2001), the prospect of increasing restrictions on copper-based biocides (Dafforn et al. 2011), and regulatory obstacles to the approval of new biocidal compounds. Two alternative strategies to biocidal coatings are non-toxic AF surfaces which rely on surface texture or surface chemistry to deter settlement of fouling organisms, and 'fouling-release' (FR) coatings where settlement and growth of fouling organisms is not necessarily prevented, but the resulting adhesion strength is very low (Lejars et al. 2012). Accumulated biofouling can therefore be released easily by hydrodynamic forces generated by vessel activity or through manual cleaning. However, current commercially available FR coatings do not prevent the formation of diatom slimes (Holland et al. 2004), are relatively soft and easily damaged, and are effective only when the coated object is in motion above a certain minimum speed (Yebra et al. 2004).

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 Supplemental data for this article can be accessed here. <https://doi.org/10.1080/08927014.2017.1383983>

One promising non-toxic AF approach is based on hydrophilic coatings that prevent adherence of marine organisms. It is well known that surfaces functionalised with poly(ethylene glycol) (PEG) are resistant to protein adsorption (Zhang et al. 2001; Ma et al. 2004; Ekblad et al. 2008; Krishnan et al. 2008; Leng et al. 2015), and their resistance to the settlement of marine fouling organisms has been demonstrated (Ekblad et al. 2008). However, PEG is readily subject to oxidation, especially in the presence of oxygen and transition-metal ions, and loses its function in biological media (Ostuni et al. 2001; Chen et al. 2005), reducing its utility for a wide range of long-term applications. An alternative emerging class of non-toxic hydrophilic antifouling coatings is based upon zwitterionic polymers. It has been demonstrated that zwitterionic polymers are highly resistant to protein adsorption and cell adhesion (Chen et al. 2005; Zhang et al. 2006; Cheng et al. 2008; Yin et al. 2013) and it is generally believed that water plays an important role in their resistance to fouling. Zwitterionic materials, containing both positive and negative charged units, possess the ability to strongly bind water molecules via electrostatic interactions, forming a hydration layer without disrupting the H-bonding interactions between water molecules (Kitano et al. 2005; Chen et al. 2005, 2010; Schlenoff 2014). Their excellent resistance to protein adsorption and to cell adhesion make zwitterionic coatings promising candidates to deter the formation of the protein film which is considered to represent the first stage in the biofouling process. Since encouraging early laboratory studies (Zhang et al. 2009) with zwitterionic polymer brushes, polymer coatings displaying zwitterions have been prepared and their anti-biofouling performance investigated in laboratory-based assays against a selection of organisms. Polyurethane-based coatings incorporating poly(sulfobetaine methacrylate)-containing triblock copolymers showed improved fouling-release performance in laboratory assays with the diatom *Navicula incerta* and the bacterium *Halomonas pacifica* (Bodkhe et al. 2014). Polysulfone and polyacrylate-based zwitterionic coatings showed good FR performance against *N. incerta*

(Hibbs et al. 2015). Zwitterionic polymer surfaces based on sulfobetaine and carboxybetaine methacrylates also appeared to alter exploration behaviour of cypris larvae of *Balanus amphitrite*, and ultimately deterred settlement of this barnacle species (Aldred et al. 2010). These encouraging results suggest that zwitterion-containing coatings are worthy of further study.

This study investigates the hypothesis that the incorporation of relatively small quantities of the zwitterionic monomers sulfobetaine methacrylate (SBMA) and carboxybetaine methacrylate (CBMA) (Figure 1) into a coating may improve its AF performance against the diatom *N. incerta*, a common fouling microalga contributing to so-called 'slime layers'. The test coatings were based upon lauryl methacrylate, which is comparatively hydrophobic; diatoms such as *N. incerta* are known to favour more hydrophobic surfaces (Holland et al. 2004), thus making it easier to observe any potential enhancements in anti-biofouling performance afforded by the incorporation of zwitterions. On account of the significant financial costs associated with zwitterionic monomers, it is arguable that a zwitterionic marine anti-biofouling coating would only be commercially viable if its zwitterionic content were relatively low, and thus only coatings containing 0–20% zwitterion on a molar basis were tested. The effects on fouling by *N. incerta* of systematically altering the balance of positively and negatively charged monomers in LMA-based coatings was also investigated by incorporating mixtures of the positively charged monomer [2-(methacryloyloxy)ethyl]trimethylammonium chloride (TMAMA) and the negatively charged monomer 3-sulfopropyl methacrylate (SMA) (Figure 1). These monomers are considerably lower in cost than betaines, and thus could provide a 'pseudo-zwitterionic' nature to coatings in a more cost-effective manner. Tetra(ethylene glycol)-di-methyl methacrylate (TEGDMA) was included as a cross-linker in all coatings to aid their stabilities. Atomic force microscopy (AFM) characterisation of the coatings was also performed to determine surface topologies and Young's modulus, two properties which may also influence the degree of fouling.

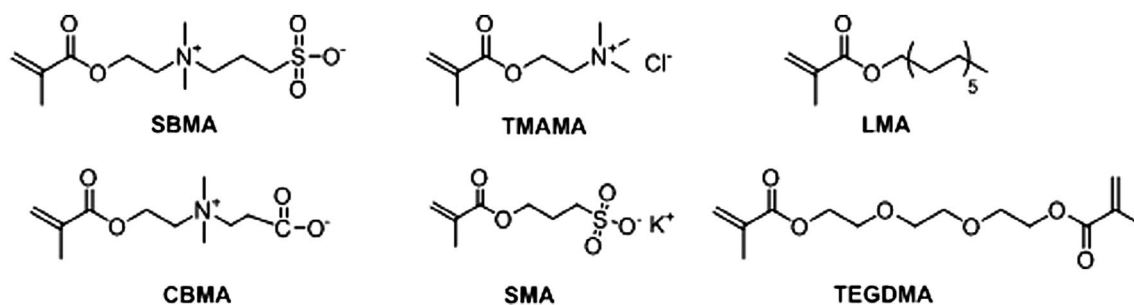


Figure 1. Chemical structures of the monomers and cross-linker used for the preparation of the coatings.

Materials and methods

Absolute EtOH was purchased from Fisher Scientific (Loughborough, UK). All other chemicals were obtained from the Sigma-Aldrich Company (Gillingham, UK). Artificial seawater at 32 psu was prepared using Tropic Marin® Sea Water Classic (Wartenberg, Germany).

Surface preparation

Prior to use, glass microscope slides (VWR Ltd, Lutterworth, UK) were washed with detergent, rinsed with deionised (DI) water, sonicated in DI water for 15 min, sonicated in EtOH for 15 min, dried with a stream of N₂ gas and subjected to oxygen plasma at 70 W for 15 min in a plasma asher. Methacryloxypropyltrimethoxysilane-modified glass substrata were prepared by submersion in a solution of γ -methacryloxypropyltrimethoxysilane (MPS) in EtOH (0.07 M) and covered. After 16 h, the substrata were removed from the bath and dried with a stream of N₂ gas. The silane layer was then cured by placing the substrata in an oven at 115°C for 30 min. Once cooled, the substrata were sonicated in EtOH to remove any silane multilayers and then dried with nitrogen. The coatings were prepared by polymerising them directly onto the functionalised substrata. A sandwich assembly of a clean glass substratum (top) and a functionalised glass substratum (bottom) was assembled, using 0.31 mm gauge syringe needles as spacers and clips as holders (Figure S1).

Coating preparation

The mixtures of monomers (16 mmol) in appropriate ratios were dissolved in a mixture of hexyl acetate/methanol (1:1, 2 ml) and stirred for 20 min. TEGDMA (640 μ l, 2.44 mmol) and lauryl peroxide (LPO) (400 mg, 1 mmol) were mixed with the solution and stirred vigorously for 10 min. Then, 144 μ l of 4-*N*, *N*-trimethylaniline (TMA) were added to the mixture and stirred for a further 1 min. The monomer formulations were injected into a sandwich of two glass microscope slides (Figure S1) separated by 0.31 mm gauge needles and the polymer coating was formed after 48 h at room temperature. The lower of the glass microscope slides was previously silanised (Scheme 1) to display methyl methacrylate functionalities capable of copolymerising with the formulation in order to covalently bond the coating to the glass substratum. The 'sandwich' was soaked in a salt water bath for two to four days to remove the non-functionalised top glass substratum. The samples were then placed in a salt water bath to leach residual impurities using fresh salt water daily for one week. Coatings were prone to cracking when allowed to dry in air for several days.

Contact angle

Contact angles (CA, θ) were measured using a KSV Cam101 (KSV Instruments Ltd, Finland) at room temperature using the captive bubble method, with air as the light phase and DI water as the heavy phase. All measurements were replicated three times and expressed as a mean, with variance expressed as $\pm$ one standard deviation (SD) from the mean.

Fourier transform infrared spectroscopy (FTIR)

FTIR spectra (in the range of 400–4,000 cm⁻¹ wave-numbers) were recorded using the ATR accessory of an IRAffinity-1S Fourier transform infrared spectrophotometer at 4 cm⁻¹ spectral resolution. For each spectrum 64 scans were co-added and averaged. The sample was measured as a dry polymer film and the bare ATR accessory was used as a background. Spectral references (sulfobetaine) were obtained from the AIST Spectral Database for Organic Compounds (http://sdb.sdb.aist.go.jp/sdb/cgi-bin/cre_index.cgi).

Atomic force microscopy (AFM)

A Digital Instruments Multimode-8 with Nanoscope V controller and E scanners (Bruker, Germany) was used for acquiring AFM images and force curves. Nanoscope software version 9.1 was used to control the microscope. The AFM data were processed with NanoScope Analysis 1.50 software (Bruker, Billerica, USA). All experiments were performed in an AFM fluid cell and under artificial seawater to investigate surface topology, adhesion and Young's modulus of the test coatings. Silicon tips on V-shaped Si₃N₄ cantilevers (model DNP, Bruker) were used for imaging and force measurement. The AFM probes were cleaned by exposure to oxygen plasma at 70 W for 10 min. For morphology and roughness measurements, ScanAsyst mode in fluid with low load force (of the order of piconewtons) was used to minimise deformation or damage to the samples. Roughness was measured on a sample area of 100 μ m². The root mean square of roughness (R_q) was calculated from five randomly selected areas of each coating. Force volume–contact mode was employed to obtain force curves that were used to assess Young's modulus and adhesion force. For both Young's modulus and adhesion forces, the deflection sensitivity of each cantilever was calibrated by performing closed-loop Z force curves on a flat array of silicon. A titanium sample with very sharp grain features (RS-12M, Bruker) was used to measure the radius of curvature of the AFM probes. The spring constant of each cantilever was calibrated using the thermal tune method (Ohler 2007).

Diatom adhesion/ease-of-removal assay

Samples were fully equilibrated in artificial seawater (30 psu) prior to testing. Cells of the laboratory-cultured diatom, *N. incerta*, were resuspended in 0.22 µm filtered artificial seawater (Tropic Marin, 30 psu). As a reference surface, polydimethyl siloxane (PDMS; clear Dow Corning 3-0213, Dow Corning Corporation, Auburn, USA) was cast onto glass slides and pre-leached in filtered de-ionised water for 28 days, and transferred to artificial seawater (30 psu) for 24 h prior to testing. Five replicate slides of each surface were placed in quadriPERM dishes (Greiner, Stonehouse, UK) and 10 ml of diatom suspension were added to each slide well. Slides were left in natural light at room temperature for 2 h to settle, and then gently washed in artificial seawater to remove unattached cells. Two replicates of each surface were exposed to a shear stress of 25.76 Pa in a flow cell for 5 min (Schultz et al. 2000). Slides were examined using fluorescence microscopy, with illumination at 546 nm (excitation) / 590 nm (emission). For each replicate slide, the density of diatom cells was measured as the average of 12 manual counts of cells per field of view. The percentage removal of diatom cells following shear exposure was calculated for each replicate slide compared to the mean density of diatom cells on slides not exposed to shear. Diatom densities with and without shear exposure, and the percentage removal of cells following flow exposure, were compared among surfaces using ANOVA, with Welch's correction where variances were unequal.

Biofilm growth/removal test

Biofilms were developed on the SBMA- and CBMA-based coatings at a 'slime farm' facility. This consisted of a 120 × 165 × 30 cm flow-through seawater tank fed with sand-filtered natural seawater maintained at 19.5 ± 1°C, 35 ± 1 psu, and pH 7.8. Samples, along with slides coated with grey-pigmented PDMS (Fe₂O₃ and TiO₂ pigments)

were placed in quadriPERM dishes at the bottom of the tank for seven weeks to allow biofilm development. Slides were photographed after two, four and seven weeks. After seven weeks, samples were tested for biofilm release. Biofilmed slides were mounted in a flow cell and fully turbulent flow was passed across the surfaces. Flow speed was increased step-wise from zero to 2.4 m s⁻¹, with speed kept constant for 60 s at each speed (speed increments were 1.4, 1.7, 2.0, 2.4 m s⁻¹). Slides were photographed before each speed increase, recording the speed at which all visible biofilm was removed.

Results and discussion

Synthesis of zwitterionic coatings

A simple route was developed for preparing zwitterionic and pseudo-zwitterionic hydrogel coatings (C1–C12) on standard glass microscope slides. Mixtures of monomer formulations (Table 1), which utilise a redox initiator system LPO/TMA, were injected into a sandwich of two glass microscope slides separated by 0.31 mm gauge needles and the polymer coating was obtained after 48 h. The lower of the glass microscope slides was silanised (Figure 2) to display methyl methacrylate functionalities capable of copolymerising with the formulation in order to covalently bond the coating to the glass substratum. After polymerisation, the sandwich system was soaked in artificial seawater for two to three days in order to remove the top glass, and the coatings were then immersed in fresh artificial seawater for one week to leach unreacted monomers. After leaching coating thickness was found to be in the range ~0.1–0.3 mm.

Physico-chemical characterisation

To verify that the composition of the polymer coatings reflected monomer and cross-linker feed ratios, elemental analysis of each coating was performed. In all cases

Table 1. Summary of the zwitterionic polymer coatings prepared.

Coating	Content of SBMA (%) ^(a)	Content of CBMA (%) ^(a)	Content of SMA (%) ^(a)	Content of TMAMA (%) ^(a)	Content of LMA (%) ^(a)	Air in water contact angle (°)
C1	0	0	–	–	100	43 ± 3
C2	1	–	–	–	99	41 ± 2
C3	10	–	–	–	90	37 ± 6
C4	20	–	–	–	80	28 ± 3
C5	–	1	–	–	99	42 ± 5
C6	–	10	–	–	90	34 ± 4
C7	–	20	–	–	80	27 ± 5
C8	–	–	10	0	90	32 ± 3
C9	–	–	7.5	2.5	90	36 ± 4
C10	–	–	5	5	90	39 ± 3
C11	–	–	2.5	7.5	90	40 ± 1
C12	–	–	0	10	90	52 ± 1

^aPercentage composition on a molar basis.

the measured elemental compositions matched within experimental error the expected compositions (Table S1 in Supporting information), suggesting that all monomers and crosslinkers were successfully incorporated into the coatings. To further confirm the coating compositions, FTIR analysis of the coatings was performed. The FTIR spectra of zwitterionic coatings comprising LMA and SBMA (C1–C4) were dominated by the LMA signals although diagnostic sulfonate-related bands were visible near $1,040\text{ cm}^{-1}$ (Figure S2 in Supporting information). The 10% SBMA (C3) and 20% SBMA (C4) samples were distinguishable by FTIR using this band, and integration of the spectra across the band affords ratios of 1.8:1, in satisfactory agreement with the expected 20:10 monomer ratios based upon monomer compositions. It was not possible to distinguish the 1% SBMA containing sample (C2) from pure poly(LMA) (C1) because of the overlap of S–O stretching modes with other dominant C–H bending/deformation and C–N modes in the $1,100\text{--}1,500\text{ cm}^{-1}$ region. For the pseudo zwitterionic coatings C8–C10 it was possible to use the sulfonate-related bands visible

near $1,060\text{ cm}^{-1}$ to verify the content of SMA (Figure S3 in Supporting information). It was not possible to use either FTIR or Raman spectroscopy to characterise the coatings containing CBMA (C5–C7) as diagnostic signals were too similar to distinguish even at 20% CBMA. FTIR spectra of coatings comprising LMA (90%) and SMA and TMAMA comonomers (10%) were dominated by the signals for LMA, but sulfonate-related bands are visible near $1,060\text{ cm}^{-1}$ (Figure S3 in Supporting information, black rectangle). Integration of this band allows the relative compositions of the coatings containing 10% SMA (C8), 7.5% SMA (C9) and 5% SMA (C10) to be confirmed. Coatings containing smaller fractions of SMA gave spectra that were difficult to distinguish from LMA. Taken together, the results of elemental analysis and FTIR studies indicate that the compositions of the polymer coatings reflects their monomer and crosslinker feed ratios on a relative basis.

The polymer coatings were characterised by measuring the contact angles (Table 1, Figure 3). The coating containing 0% zwitterion (C1) had the highest contact

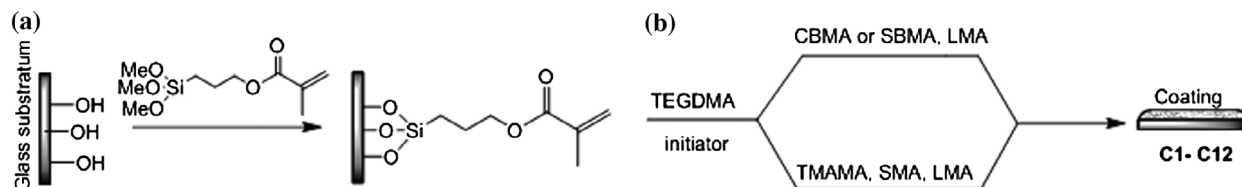


Figure 2. Preparation of polymer coatings on glass substrata using LMA, TEGDMA and the zwitterionic monomers SBMA or CBMA, or a combination of the positively and negatively charged monomers SMA and TMAMA. Reaction conditions: (a) EtOH, room temperature, 16 h; (b) LPO/TMA, MeOH/hexyl acetate, room temperature, 48 h.

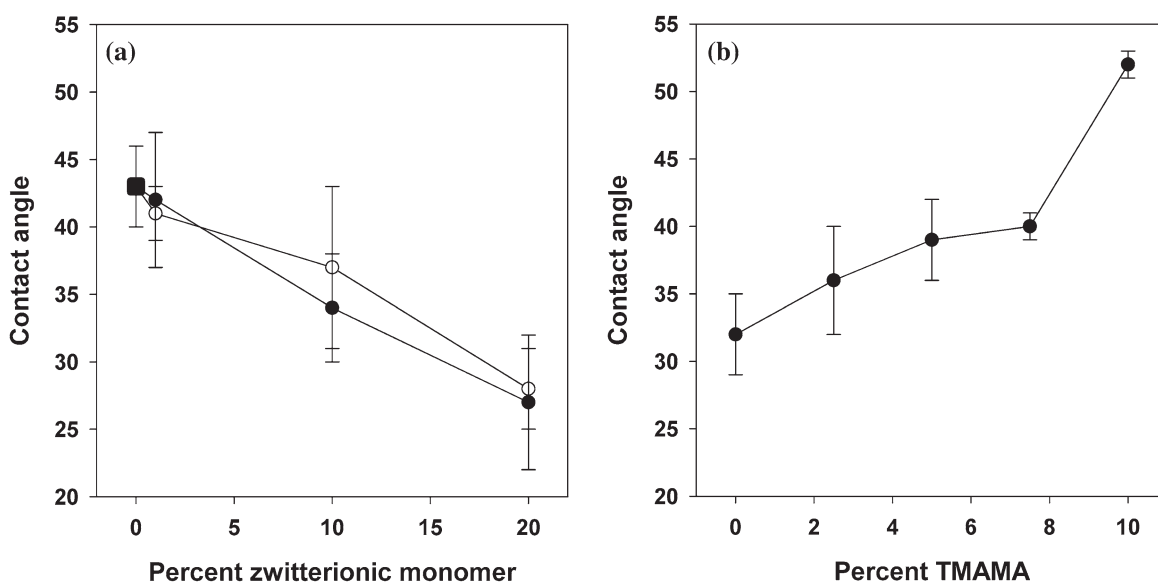


Figure 3. (a) Relationship between contact angles ($^{\circ}$) and the percent content of the zwitterionic monomers SBMA (white symbols, surfaces C2–C4) and CBMA (black circles, surfaces C5–C7). The black square is pure LMA (C1). (b) Relationship between contact angles ($^{\circ}$) and the percent content of TMAMA in coatings C8–C12. Error bars are ± 1 SD.

angle, $43 \pm 3^\circ$, indicating it possessed the lowest hydrophilicity. Contact angles decreased as the content of zwitterionic monomer increased and similar values were observed between the SBMA and CBMA series (Figure 3A). This observation indicates that coating hydrophilicity increased as zwitterion incorporation increased, and suggests that zwitterionic monomers were displayed upon the surfaces. For coatings C8–C12, contact angle increased with the content of positively charged monomer (Figure 3B), indicating a concurrent decrease in coating hydrophilicity. Contact angles obtained for these coatings were largely in agreement with other hydrogel-like coatings reported in the literature (Magin et al. 2011; Yin et al. 2013; Paradiso et al. 2014). The water content and swelling ratios (Q) of coatings were also determined (Table S1 in Supporting information).

AFM topographical imaging of all coatings was performed on a sample area of $100 \mu\text{m}^2$ to provide information on surface topology (Figure 4, Table 2). Coating C1 (0% zwitterion) was relatively flat and featureless (Figure 4A) with a root mean square of roughness (R_q) of 11 nm (Table 2). Coating C2 (1% SBMA) displayed (Figure 4B) irregular and μm -sized dimple-like features within an otherwise relatively flat surface. Coating C3 (10% SBMA) displayed a higher density of smaller dimple-like features

(Figure 4C), whereas coating C4 (20% SBMA) appeared to have lost the dimple-like effects, instead presenting an uneven surface topology (Figure 4D). These observations suggest that incorporating increasing amounts of SBMA into the LMA-based coating led to considerable increases in surface roughness with significant and unpredicted differences in surface topologies. The reasons for this change in topology are currently unclear but most likely arise because of a degree of phase separation when SBMA monomers are formulated into the hydrophobic LMA-based coating. The incorporation of CBMA into the LMA-based coating, on the other hand, did not lead to the formation of any significant surface features (Figure 4E–G), with only small increases in surface roughness observed as increasing quantities of CBMA were incorporated (Table 2). These observations suggest that incorporating CBMA into the LMA-based coating resulted in less disruption to the surface topology than the addition of SBMA. For the coatings (C8–C12), AFM analysis revealed relatively flat and featureless surfaces throughout the series (Figure 4H–L), and the measured surface roughness was found to be within a relatively narrow range of ~ 12 –44 nm (Table 2). These observations suggest that incorporating TMAMA and/or SMA into the LMA-based coating also did not have a substantial effect on surface topology or

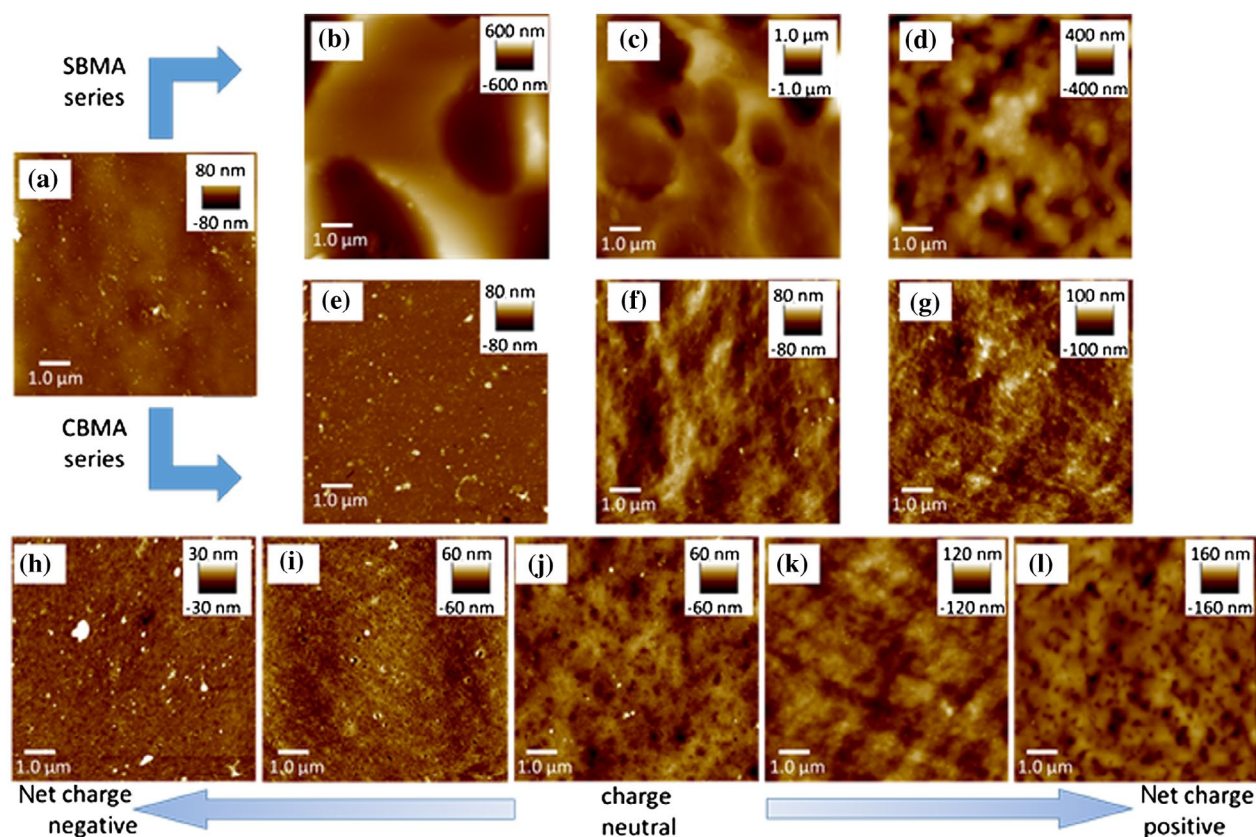
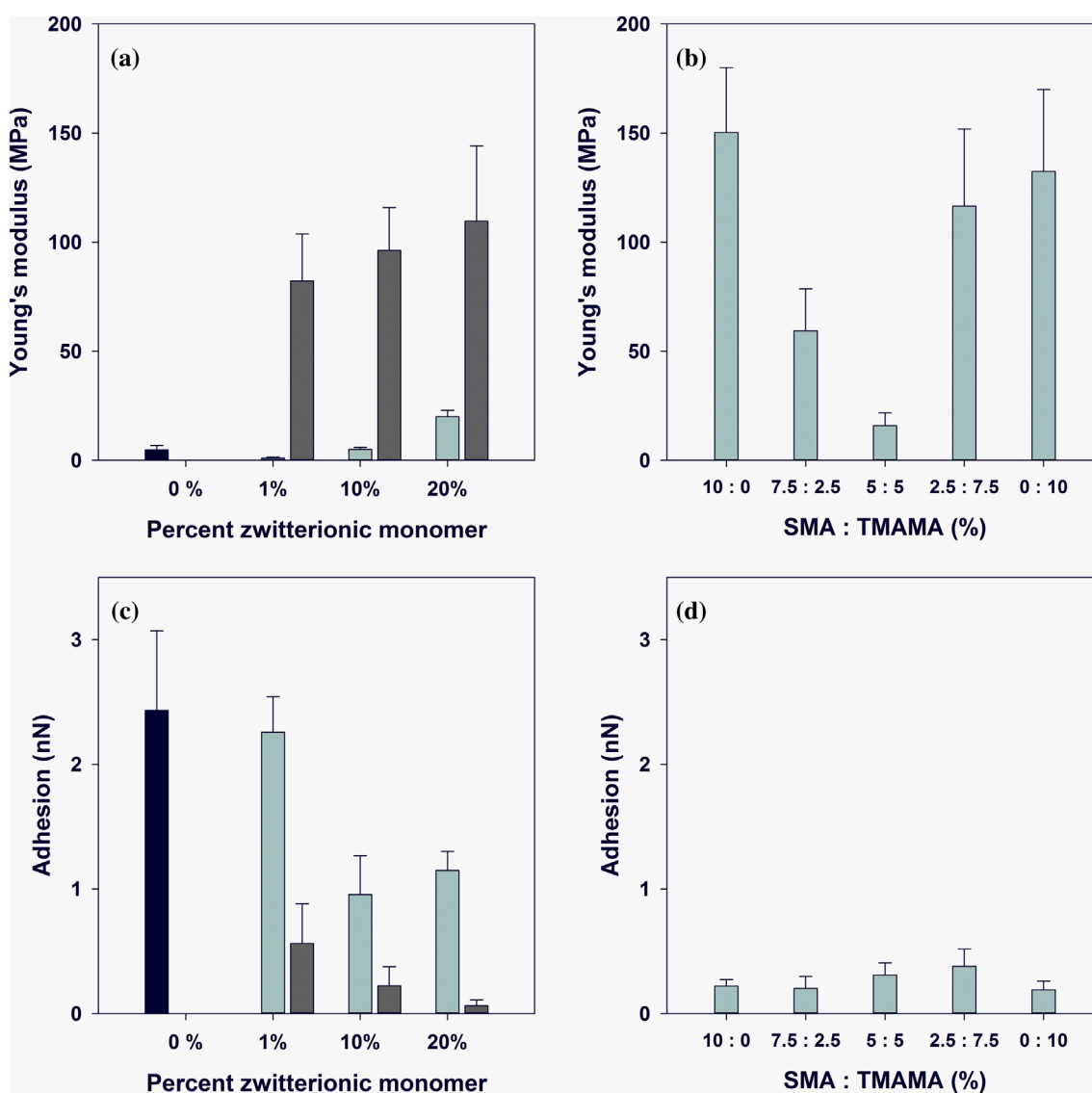


Figure 4. Representative AFM topographical images of (a) the LMA coating (C1); (b–d) SBMA-containing coatings (C2–C4); (e–g) CBMA-containing coatings (C5–C7); and (h–l) the mixed-monomer coatings (C8–C12).

Table 2. Root mean square of roughness (R_q), Young's modulus and the adhesion force of the polymer coatings.

Coating	Roughness (nm)	Young's modulus (MPa)	Adhesion (nN)
C1	10.76	4.83	2.43
C2	153.80	0.98	2.26
C3	159.20	4.94	0.95
C4	105.74	20.03	1.15
C5	11.09	82.22	0.56
C6	19.18	96.16	0.22
C7	24.88	109.65	0.06
C8	11.87	150.30	0.22
C9	21.56	59.33	0.20
C10	14.04	15.89	0.31
C11	28.66	116.50	0.38
C12	44.10	132.38	0.19

roughness. AFM nano-indentation experiments were also performed to determine Young's modulus, which provides a measurement of coating stiffness. This experiment also allowed determination of adhesion, a measurement of the adherence between the AFM probe and the surface. Addition of CBMA resulted in a large increase in Young's modulus, with greater proportions of CBMA giving higher values (Figure 5A, Table 2). A decrease was observed when 1% SBMA was incorporated (C2; Figure 5A, Table 2), but addition of greater proportions of SBMA appeared to lead to a modest increase in Young's modulus (C3–C4). The results suggest that SBMA-containing coatings are, in general, more stiff and rougher than their

**Figure 5.** (a) Young's modulus of LMA coating C1 (black) and as a function of SBMA content (light grey) and CBMA content (dark grey). (b) Young's modulus as a function of TMAMA content in surfaces C8–C12. (c) Adhesion strength of AFM tip to LMA coating C1 (black) and as a function of SBMA content (light grey) and CBMA content (dark grey). (d) Adhesion strength of AFM tip as a function of TMAMA content in surfaces C8–C12. Error bars are +1 SD.

CBMA-containing or mixed-charge counterparts. The Young's moduli of coatings (C8–C12) are shown in Figure 5B and Table 2, where it can be seen that the minimum value is obtained when the charged monomers are present in near equimolar ratios. Those coatings possessing excess positively or negatively charged monomer compositions display larger values of Young's modulus.

Surface adhesion decreased as SBMA or CBMA were formulated into the coatings (Figure 5C, Table 2). For the series C8–C12 (Figure 5D, Table 2), adhesion was fairly constant within the range 0.20–0.38 nN. These observations suggest that the introduction of zwitterionic monomers reduced adhesion strength between the AFM tip and the surface. Taken together, these AFM studies highlight that the incorporation of small amounts of zwitterion into LMA-based coatings can have significant effects upon the resulting topology and mechanical properties, even at only 1% zwitterion.

Diatom adhesion/ease-of-release

The AF behaviour of the coatings was assessed in the laboratory against the diatom *N. incerta*. Compared to pure LMA, coatings containing SBMA (C2–C4) had similar densities of adhered diatom cells prior to exposure to shear stress (Welch's ANOVA; $F = 1.21$, $df = 3$, $p = 0.41$). However, diatom density after shear exposure did differ among coatings with different levels of SBMA (ANOVA; $F = 7.35$, $df = 3$, $p = 0.042$); the density of cells was greatest on the surface with the highest proportion of SBMA (C4; Figure 6A). Incorporation of SBMA into LMA also had a significant effect on percent removal of cells (ANOVA; $F = 7.0236$, $df = 3$, $p = 0.045$), with lowest proportional removal from the surface with most SBMA (C4; Figure 6B). These data suggest that incorporation of increasing amounts of SBMA decreased AF performance relative to pure LMA.

There was no significant variation in the diatom cell density among the CBMA series (C5–C7) prior to exposure to shear (Welch's ANOVA, $F = 0.7526$, $df = 3$, $p = 0.58$). Inclusion of increasing amounts of CBMA had an effect on both final cell density (ANOVA; $F = 10.87$, $df = 3$, $p = 0.022$) and percentage removal (ANOVA, $F = 9.046$, $df = 3$, $p = 0.030$) of *N. incerta* cells. In contrast to SBMA, CBMA improved the performance of LMA, with the 20% CBMA (C7) surface having the lowest final cell density (Figure 6C) and the highest percent removal (Figure 6D). These data suggest that, in contrast to SBMA, incorporation of CBMA did improve the AF performance of the underlying LMA coating. This differs from the results of other studies, which have found that pure SBMA self-assembled-monolayers (SAMs) performed

better than similar CBMA surfaces in AF assays using the same diatom species (Finlay et al. 2013; Bauer et al. 2016).

It may seem surprising that the zwitterionic surfaces did not perform better than PDMS, since previous studies of zwitterionic surfaces generally suggest superior performance against diatoms (Bodkhe et al. 2014; Hibbs et al. 2015). The reason for this discrepancy is likely to result from the underlying properties of the LMA surface; the AF performance of 100% LMA is similar, to PDMS (Figure 6; LMA is coating C1). The addition of the zwitterionic material to the LMA coatings, at the levels tested here, appears to have had only a small impact on the AF performance of LMA.

Varying the proportion of two oppositely charged monomers (C8–C12) had no significant effect on the initial densities of adhered diatom cells (ANOVA; $F = 0.357$, $df = 4$, $p = 0.833$), the density of cells after exposure to shear stress (ANOVA; $F = 1.025$, $df = 4$, $p = 0.465$), or the percentage of cells removed following exposure to shear stress (ANOVA; $F = 0.949$, $df = 4$, $p = 0.497$). Though all differences among test surfaces were non-significant, it did appear (Figure 6E and F) that the worst performing surface (highest final density, lowest percentage removal) was the surface with equal amounts of the two charged monomers (C10). None of the surfaces performed better than unmodified LMA (C1). It is worthwhile noting that there does not appear to be any correlation between the water content and swelling ratios (Q) of coatings and their AF performance.

Biofilm growth/removal test

The test surfaces showed very low fouling even after seven weeks immersion in the 'slime farm' tank, with all the LMA-based surfaces appearing to show less fouling than the PDMS standards (Figure 7). Coatings C3 and C5–C7 underwent biofilm removal testing in the flow cell; the remaining coatings only showed a low amount of weakly adhered biofouling that was removed during transportation. At a flow speed of 1.7 m s^{-1} all visible biofilm was removed from C5 and C7 (Figures S4–S6 in Supporting information), while full removal from C3 and C6 required the slightly higher flow speed of 2.0 m s^{-1} . The PDMS standards were still visibly biofilmed after exposure to the maximum flow speed of 2.4 m s^{-1} (Figure S8 in Supporting information). These results indicate generally good AF and FR performance against the biofilms formed in the 'slime farm'. In contrast to the diatom removal assay results, they also show a markedly better performance than the PDMS standards. However, there was little detectable variation among different levels of the two different zwitterions.

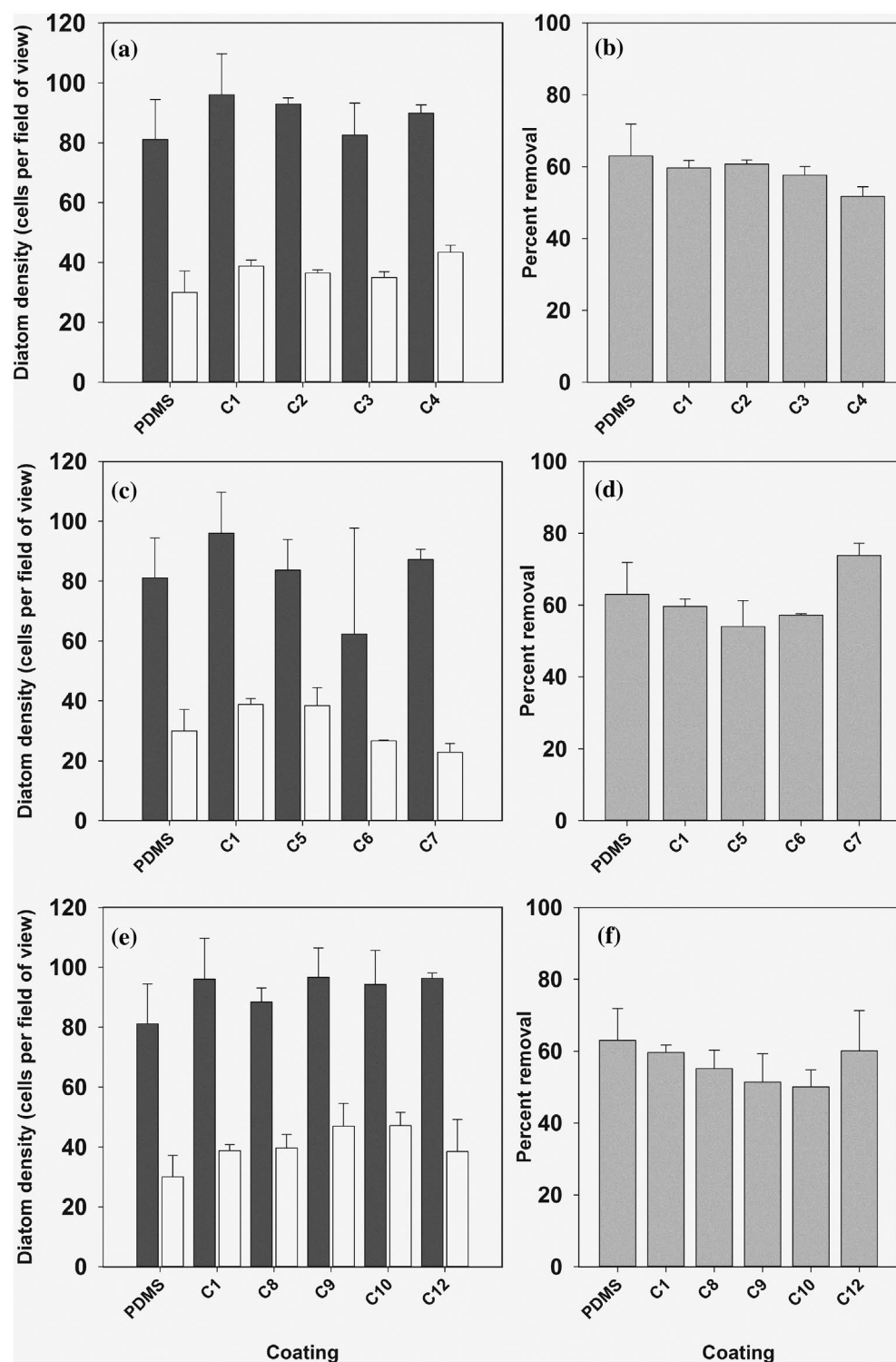


Figure 6. Performance of test coatings in adhesion and ease-of-removal assay using *N. incerta*. Dark grey bars show density of cells with no exposure to shear; white bars show density after exposure to a shear stress of 25.76 Pa for 5 min (a, c, e). Light grey bars show the percent removal of cells resulting from exposure to shear (b, d, f). (a, b) SBMA series: C1 – 100% LMA; C2 – LMA with 1% SBMA; C3 – LMA with 10% SBMA; C4 – LMA with 20% SBMA. (c, d) CBMA series: C1 – 100% LMA; C5 – LMA with 1% CBMA; C6 – LMA with 10% CBMA; C7 – LMA with 20% CBMA. (e, f) Mixed monomer surface series (for detailed composition data see Table 1); no data were obtained for surface C11. Error bars are + 1 SD.



Figure 7. Coatings C1–C7 and PDMS standards after seawater immersion in the ‘slime farm’ for two weeks (left) and seven weeks (right). Slides had not been exposed to shear.

Effectiveness of SBMA and CBMA as additives to the LMA polymer surface

The differences observed in AF performance against *N. incerta* between the coatings containing SBMA and those containing CBMA may arise from differences in how the monomers interact with water, differences in surface properties, or combinations of these two factors. Regarding their interactions with water, carboxybetaines have cationic and anionic groups that are quite different in charge densities, leading to fewer associations between monomers (Shao et al. 2014) and consequently moieties in carboxybetaine materials are very hydrated. This feature is important, as the fouling resistance properties of zwitterionic materials may arise from the formation of a strong hydration layer without disturbing the H-bonding interactions between water molecules (Chen et al. 2005, 2010; Kitano et al. 2005; Schlenoff 2014). Sulfobetaines possess more similar charge densities, and being less well hydrated might be less effective in anti-biofouling. Regarding possible differences in surface properties, the incorporation of SBMA did reduce the adhesion between the AFM tip and the coatings. However, any beneficial effect of the presence of SBMA on surface chemistry may have been masked by the effect of SBMA on coating roughness. Roughness at various scales has a strong influence on AF performance (Schumacher et al. 2007; Scardino et al. 2009) and it is possible that the lower performance of the SBMA-containing coatings was driven by the uncontrolled and substantial increases in surface roughness in the coatings. Inclusion of CBMA, on the other hand, did not affect surface roughness to the same degree, although it did lead to increased Young’s modulus and reduced AFM tip adhesion compared to both unmodified LMA and the SBMA-modified coatings. These differences in the effects of the two zwitterions on surface properties would appear to explain why the results are

different from similar experiments conducted using pure zwitterionic SAMs (Finlay et al. 2013; Bauer et al. 2016). The results of the biofilm assay do not appear to differentiate between coatings with SBMA and CBMA, and show generally good performance on all LMA-based coatings. It is possible that this test was less sensitive to differences among these particular coatings.

The coatings C8–C12 did not vary significantly in performance, which is in agreement with some of the physical characterisation data; contact angle, roughness, and AFM tip adhesion did not vary substantially among these coatings. There did appear to be a (non-significant) tendency for the surfaces with a net excess of positively or negatively charged monomers to perform better than the surfaces with a more balanced ratio of negative and positive charges. This behaviour was not anticipated since it has been reported that coatings formed by mixed positively and negatively charged self-assembled monolayers (SAMs) or hydrogel of equal valence are highly resistant to protein adsorption (Holmlin et al. 2001; Chen et al. 2006; Chen and Jiang 2008). Young’s modulus varied substantially among the coatings C8–C12, and this appeared to reflect differences in coating performance – the worst performing coating (C10) was the one with the lowest Young’s modulus, while the coatings with the highest Young’s modulus (C8 and C12), performed best.

Conclusions

Polymer surfaces incorporating relatively low amounts of zwitterion have been prepared, characterised and their performance assessed in the laboratory against *N. incerta*. These results suggest that incorporation of SBMA into LMA-based coatings did not enhance the FR properties, possibly on account of the nature of the monomer and/or changes in surface roughness, but that the incorporation of CBMA did

lead to an improvement. Coatings containing a mixture of positively and negatively charged monomers performed best when there was a net excess of either positively or negatively charged monomers, rather than when there were equimolar amounts of positively and negatively charged monomers. All coatings performed well in controlled-field studies where they were immersed in filtered seawater for several weeks, with all fouling being removed easily with relatively low hydrodynamic shear. The results suggest that the incorporation of relatively small amounts of zwitterions into polymer coatings may lead to enhancements in FR properties, although much work will have to be done to identify optimal formulations, with particular attention paid to the possible effects of surface roughness and topology.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This study received funding from the European Community's Seventh Framework programme FP7/KBBE [grant number 614034] (SEAFRONT).

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